Ehtesham Aalam

B.Tech. in Electrical Engineering | NIT Patna
BS in Data Science & Applications | IIT Madras
Batch of 2024 – 2028

+91-7705880750
alamehtesham88@gmail.com
linkedin.com/in/ehtesham-aalam
github.com/23f2001511

Education

Degree / Institute	CGPA / %	Year
B.Tech. in Electrical Engineering National Institute of Technology, Patna	7.78/10	2024–2028
BS in Data Science and Applications Indian Institute of Technology, Madras		2023–2028
Class XII (Senior Secondary) UP Board, MAA K G I C Medanipur	84.0%	2023
Class X (Secondary) UP Board, MAA K G I C Medanipur	85.8%	2021

Technical Skills

- **Programming Languages:** C++, Python, Java (Core/OOP), SQL, JavaScript, HTML5, CSS3
- **Web Development:** Flask, Jinja2, Bootstrap, RESTful APIs, MVC pattern
- **Databases & Tools:** PostgreSQL, SQLite, SQLAlchemy, Git, GitHub, VS Code, Linux, MATLAB
- **Core CS Concepts:** Data Structures & Algorithms (DSA), OOP, DBMS (joins, normalization)

Projects

- **Hospital Management System – Web Application** *Jan 2025 – Apr 2025*
  Tools: Python, Flask, PostgreSQL, SQLAlchemy, Jinja2, Bootstrap, HTML5, CSS3 GitHub
  – Developed a full-stack web application with role-based access control for Admin, Doctor, and Patient portals.
  – Designed an 8-table relational schema in PostgreSQL using SQLAlchemy ORM to organize application data.
  – Built responsive dashboards for appointment booking and patient histories, with input validation and an MVC structure.

- **Clap-Activated Electrical Appliance Control System** *Nov 2024*
  Tools: Analog Electronics, IC 555 Timer, Transistor Circuitry, 5V Relay, Condenser Microphone
  – Built a sound-triggered switching circuit for AC/DC appliances using a condenser microphone and transistor amplification.
  – Configured an IC 555 timer in monostable mode to reduce noise-induced false triggering.

- **Pyrolysis System: Plastic-to-Fuel Conversion Unit** *Jan 2025 – Mar 2025*
  Tools: Thermal Engineering, Experimental Design, Lab Instrumentation, Data Logging
  – Led laboratory fabrication to convert waste plastics into approximately 250 ml of combustible fuel via anaerobic thermal decomposition.
  – Designed a high-temperature heating chamber and condensation assembly, and documented operational safety procedures.

Certifications

- **IIT Madras – NPTEL,** The Joy of Computing using Python (Elite Certification, 92%) *April 2026*
- **IIT Madras BS Degree,** Foundational Level Certificate in Programming and Data Science *April 2025*
- **MathWorks Training Services,** MATLAB Onramp: Mathematical Modeling & Computation *October 2025*

Achievements

- Solved 120+ DSA problems across LeetCode and GeeksforGeeks, focusing on C++ and problem solving.
- JEE Main 2024: All India Rank 37,223 (97.63 percentile).
- Cleared the IIT Madras BS Qualifier Exam in Computational Thinking and Mathematics.

Positions of Responsibility

- **Event Lead & Core Member,** Tesla Club (Technical Society), NIT Patna *2024 – Present*
  – Organized and led **The Lost Electron**, a technical treasure hunt in September 2026, managing logistics, clue design, and event execution for 44 teams and 130+ participants.